Xinjie Liu

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Research

I develop algorithms and theory for scalable, sample-efficient, and trustworthy **robot learning**.

Examples of my work include (i) multi-fidelity methods that stabilize **real-robot reinforcement learning** by combining scarce real-world data (e.g., as few as 4 rollouts per update) with abundant but imperfect simulation data, while remaining robust to simulation bias [11, 9], and (ii) learning behavior models that predict how agents act in interaction-rich, uncertain scenarios [7, 2, 1]. I ground my research in real-world systems such as robotic manipulators and mobile robots.

Education

The University of Texas at Austin, USA — Aug 2023 – May 2028 (expected)
Ph.D. in Electrical and Computer Engineering — Advisors: Ufuk Topcu and David Fridovich-Keil

Delft University of Technology, Netherlands — Sep 2021 – Jul 2023
M.Sc. in Robotics, *cum laude*; GPA 9.04/10 (top 1.6%) — Advisor: Javier Alonso-Mora

Tongji University, China — Sep 2016 – Jul 2021
B.Eng. in Automotive Engineering; exchange semester at TU Graz, Austria (2020)

Publications

[11] **X. Liu**, R. Zhao, A. Chaudhuri, C. Neary, U. Topcu, and D. Fridovich-Keil. Multi-Fidelity Policy Gradients Stabilize Data-Scarce Reinforcement Learning, under review, 2026. Accepted at the CoRL 2026 Workshop on Roadmap to Sample-Efficient Real-World Reinforcement Learning.

[10] H. Xu, P. Srivastava, **X. Liu**, S. Agarwal, and A. Zhang. Horizon-Scalable Q-Learning, under review, 2026.

[9] **X. Liu***, C. Neary*, K. Gupta, W. A. Suttle, C. Ellis, U. Topcu, and D. Fridovich-Keil. A Multi-Fidelity Control Variate Approach for Policy Gradient Estimation, Transactions on Machine Learning Research (TMLR), 2026. Awarded **J2C Certification ($<$ 10% of accepted papers)**; selected to present at NeurIPS 2026.

[8] T. Qiu, F. Fotiadis, **X. Liu**, C. Ellis, J. Milzman, W. A. Suttle, U. Topcu, and D. Fridovich-Keil. Linear-Quadratic Gaussian Games with Distributed Sparse Estimation, IEEE Control Systems Letters (L-CSS), 2026.

[7] Y. Jain*, **X. Liu***, L. Peters*, D. Fridovich-Keil, and U. Topcu. Bayesian Inverse Games with High-Dimensional Multi-Modal Observations, under review at The International Journal of Robotics Research (IJRR), 2025.

[6] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Approximate Feedback Nash Equilibria with Sparse Inter-Agent Dependencies, conditionally accepted at IEEE Transactions on Automatic Control (T-AC), 2025.

[5] K. Gupta, **X. Liu**, R. Allen, U. Topcu, and D. Fridovich-Keil. Second-Order Algorithms for Finding Local Nash Equilibria in Zero-Sum Games, under review, 2025.

[4] C. Armstrong*, R. Park*, **X. Liu**, K. Gupta, and D. Fridovich-Keil. Inferring Foresightedness in Dynamic Noncooperative Games, IEEE Robotics and Automation Letters (RA-L), 2025.

[3] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Policies with Sparse Inter-Agent Dependencies in Dynamic Games: A Dynamic Programming Approach, Autonomous Agents and Multi-Agent Systems (AAMAS), Extended Abstract, 2025.

[2] **X. Liu***, L. Peters*, J. Alonso-Mora, U. Topcu, and D. Fridovich-Keil. Auto-Encoding Bayesian Inverse Games, Algorithmic Foundations of Robotics XVI (WAFR), 2024.

[1] **X. Liu***, L. Peters*, and J. Alonso-Mora. Learning to Play Trajectory Games against Opponents with Unknown Objectives, IEEE Robotics and Automation Letters (RA-L), 2023.

*indicates equal contribution.

Selected Honors and Awards

Journal-to-Conference (J2C) Certification, TMLR ($<10\%$ of accepted papers) 2026

Cockrell School of Engineering Fellowship, UT Austin 2023 – present

Travel Grants: Texas Robotics; WAFR 2025, 2024

Graduation *cum laude*, TU Delft 2023

First Prize, Franka Emika Robot Tactile Manipulation Challenge, European Robotics Forum 2022

First Prize Academic Scholarship, Tongji University (top 3% of department) 2019

Service and Mentoring

Conference Reviewer: NeurIPS, ICLR, RLC, ICRA, IROS, CDC
Journal Reviewer: T-RO, RA-L, T-AC

Mentees: Ryan Park, UT Austin undergraduate (next: Ph.D. at UC Berkeley)
Yash Jain, UT Austin undergraduate (next: Ph.D. at UPenn)

Skills

Programming: Python, C++, Julia
Machine Learning: PyTorch, JAX
Robotics: Isaac Lab, ManiSkill, MuJoCo, ROS, Franka arm, Jackal and Husky mobile robots
Tools: Git, Linux, SLURM
Languages: English (C1), Mandarin (native), German (B2)